

MMF062 Vehicle Motion Engineering

Design Task 2 - Group 5

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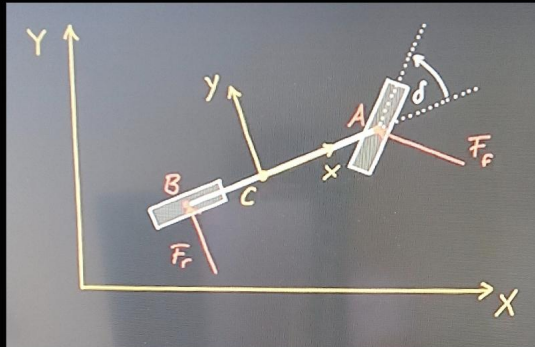
December 19, 2024

Task 1 - Steady State Cornering Characteristics

Task 1.1: Steady state cornering equations

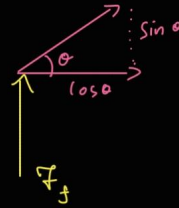
The detailed derivation is shown below.

Laws of motion



Lateral dynamics

$$m a_y = \sum_i F_{y,i}$$



$$m a_y = F_r + F_f \cos \delta$$

Here we consider only the 'y' components of the tire forces.

$$\therefore \delta \Rightarrow \text{small} \Rightarrow \cos \delta \approx 1$$

$$\therefore m a_y = F_r + F_f$$

Note :- $F_f = F_{f_{yc}} \sin(\delta) + F_{f_y} \cos(\delta)$
cause it's turning

Now since the vehicle frame is not belonging to the fixed frame, the acc of 'y' component in the fixed frame is given by

$$a_y = \dot{v}_y + \underbrace{\omega \cdot v_{yc}}_{\text{centripetal acceleration}}$$

but $\dot{v}_y = 0$ at steady state cornering!

$$\therefore m(\omega \cdot v_{yc}) = F_r + F_f \Rightarrow F_r + F_{f_y} + F_{f_z} \delta$$

yaw dynamics

by angular momentum principle,



Figure 1

Yaw dynamics

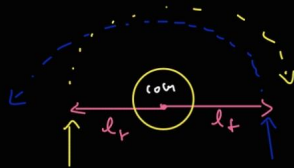
by angular momentum principle,

$$I_z \dot{\omega} = \sum_i M_i \cos \delta \rightarrow \text{sum of moments.}$$

↓
Moment of inertia about
z axis.

$$= -l_r \tau_r + l_f \tau_f \cos \delta \approx -l_r \tau_r + l_f \tau_f$$

Why $-\tau_r \neq +\tau_f$? well from CoG



lets take clockwise
as -ve $\neq$ anti-
as +ve

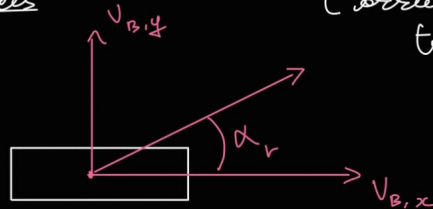
$\omega = 0$ at
steady
state

$$-l_r \tau_r + l_f \tau_f = l_f \tau_{fy} + l_f \tau_{fx} \delta - l_r \tau_{ry} = 0$$

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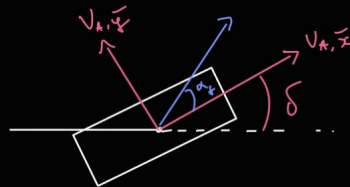
Yaw forces

Recall:-



(Assuming we follow a linear
tire model)

cont:-



Velocity geometry:-

$$\tan \alpha_r = \frac{V_{B,y}}{V_{B,x}} \approx \alpha_r \quad (\text{why? cause small slip angles})$$

Figure 2

Velocity geometry:-

$$\tan \alpha_r = \frac{V_{B,y}}{V_{B,x}} \approx \alpha_r \quad \left(\text{why? cause small slip}^* \right)$$

$$\tan \alpha_f = \frac{V_{A,y}}{V_{A,x}} \approx \alpha_f$$

linear tire model:-

$$F_f = -C_f \underline{\alpha_f} ; \quad F_v = -C_v \underline{\alpha_r}$$

Note:-

here it's negative because of the way we have considered the force direction in the F.B.D !

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Kinematic relations:-

$$V_{B,x} = V_{xc} ; \quad V_{B,y} = V_{yc} - \omega l_r$$

$$V_{A,x} = V_{xc} ; \quad V_{A,y} = V_{yc} + \omega l_f$$

$\therefore \alpha_f$ is represented w.r.t to wheel frame lets change it to the vehicle frame (i.e. $\bar{x}, \bar{y} \rightarrow x, y$)

$$\begin{bmatrix} V_{A,x} \\ V_{A,y} \end{bmatrix} = \begin{bmatrix} \cos & \sin \\ -\sin & \cos \end{bmatrix} \begin{bmatrix} V_{A,\bar{x}} \\ V_{A,\bar{y}} \end{bmatrix}$$

Figure 3

$$V_{A,x} = V_{A,\bar{x}} \cos \theta + V_{A,\bar{y}} \sin \theta$$

$$V_{A,y} = -V_{A,\bar{x}} \sin \theta + V_{A,\bar{y}} \cos \theta$$

Here $\theta = \delta$ $\Rightarrow$ we will consider our approximations!

$$V_{A,x} = V_{A,\bar{x}} + V_{A,\bar{y}} \cdot \delta$$

$$V_{A,y} = -V_{A,\bar{x}} \cdot \delta + V_{A,\bar{y}}$$

Usually they'll be:

$$V_{A,\bar{x}} \cos \delta + V_{A,\bar{y}} \sin \delta$$

$$-V_{A,\bar{x}} \sin \delta + V_{A,\bar{y}} \cos \delta$$

Now that we found everything that is needed for our tire force let's calculate them by substituting

$$F_f = -C_f \alpha_f = -C_f \cdot \frac{V_{A,y}}{V_{A,x}} = -C_f \frac{(-V_x \cdot \delta + V_{A,\bar{y}} + \omega l_f)}{V_x + (V_y + \omega l_f) \delta}$$

$$\because \delta \text{ is small, } V_x \gg (V_y + \omega l_f) \delta$$

$$F_{f_y} = C_f \delta - C_f \frac{(V_y + \omega l_f)}{V_x} \quad \text{--- (III)}$$

III $\hookrightarrow l_f$,

$$\sigma_{f_{r,y}} = -C_r \alpha_r = -C_r \frac{V_{B,y}}{V_{B,x}} = -C_r \cdot \frac{(V_y - \omega l_r)}{V_x}$$

Figure 4

||| v_y ,

$$\boxed{\ddot{y}_{ly}} = -C_r \ddot{x}_r = -C_r \frac{v_{B,y}}{v_{B,x}} = \boxed{-C_r \cdot \frac{(v_y - \omega l_r)}{v_x}}$$

(IV)

New sub eqn (III) $\oplus$ (IV) in (I) $\oplus$ (II)

$$(I) \Rightarrow m(\omega \cdot v_x) = C_f \delta - C_f \frac{(v_y + \omega l_f)}{v_x} - C_r \frac{(v_y - \omega l_r)}{v_x} + \ddot{y}_{lx} \delta$$

$$\Rightarrow m(\omega \cdot v_x) = C_f \delta - \frac{C_f v_y}{v_x} - \frac{C_f \omega l_f}{v_x} - \frac{C_r v_y}{v_x} + \frac{C_r \omega l_r}{v_x} + \ddot{y}_{lx} \delta$$

$$m(\omega \cdot v_x) + \frac{C_f \omega l_f}{v_x} - \frac{C_r \omega l_r}{v_x} = C_f \delta - (C_f + C_r) \frac{v_y}{v_x} + \ddot{y}_{lx} \delta$$

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$$\omega \left[m v_x + \frac{C_f l_f - C_r l_r}{v_x} \right] + v_y \left[\frac{C_f + C_r}{v_x} \right] = (C_f + \ddot{y}_{lx}) \delta \quad \text{--- (5)}$$

$$(II) = \left(C_f \delta - C_f \frac{(v_y + \omega l_f)}{v_x} \right) l_f - \left(- \frac{C_r (v_y - \omega l_r)}{v_x} \right) l_r + l_f \ddot{y}_{lx} \delta \quad \int = 0$$

$$\left. \begin{aligned} C_f l_f \delta - \frac{C_f l_f v_y}{v_x} - \frac{C_f \omega l_f^2}{v_x} + \frac{C_r v_y l_r}{v_x} - \frac{C_r \omega l_r^2}{v_x} + l_f \ddot{y}_{lx} \delta \end{aligned} \right\} = 0$$

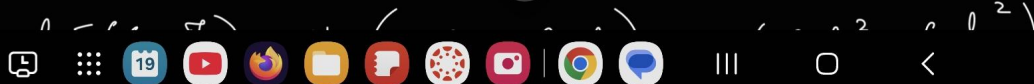


Figure 5

$$\begin{aligned}
\textcircled{II} &= \left(C_f \delta - C_f \frac{(v_y + \omega l_f)}{v_x} \right) l_f - \left(- \frac{C_r (v_y - \omega l_r)}{v_x} \right) l_r \quad \text{---} \\
&\quad + l_f F_{fx} \delta \\
&\quad \left\{ C_f l_f \delta - \frac{C_f l_f v_y}{v_x} - \frac{C_f \omega l_f^2}{v_x} + \frac{C_r v_y l_r}{v_x} - \frac{C_r \omega l_r^2}{v_x} + l_f F_{fx} \delta \right\} = 0 \\
&\quad l_f \delta (C_f + F_{fx}) + v_y \left(\frac{C_r l_r - C_f l_f}{v_x} \right) - \omega \left(\frac{C_f l_f^2 + C_r l_r^2}{v_x} \right) \\
&\quad = 0
\end{aligned}$$

$$v_y \left(\frac{C_r l_r - C_f l_f}{v_x} \right) + \omega \left(\frac{-C_f l_f^2 - C_r l_r^2}{v_x} \right) = \frac{-l_f \delta}{(C_f + F_{fx})}$$

L ⑥

Figure 6

Finally using the above equations (eqn 5 and 6) and using the matlab solver we get the equations shown below which are indeed the steady state equations.

$$v_y = \frac{\delta \cdot v_x \cdot (C_f + F_{fx}) \cdot (C_r \cdot l_r^2 + C_r \cdot l_f \cdot l_r - l_f \cdot m \cdot v_x^2)}{C_f \cdot C_r \cdot l_f^2 + 2 \cdot C_f \cdot C_r \cdot l_f \cdot l_r - C_f \cdot m \cdot l_f \cdot v_x^2 + C_f \cdot C_r \cdot l_r^2 + C_r \cdot m \cdot l_r \cdot v_x^2}$$

$$\omega = \frac{\delta \cdot v_x \cdot (C_r \cdot l_f + C_r \cdot l_r) \cdot (C_f + F_{fx})}{C_f \cdot C_r \cdot l_f^2 + 2 \cdot C_f \cdot C_r \cdot l_f \cdot l_r - C_f \cdot m \cdot l_f \cdot v_x^2 + C_f \cdot C_r \cdot l_r^2 + C_r \cdot m \cdot l_r \cdot v_x^2}$$

If it's rear wheel driven, then the F_{fx} must be set as zero. In the below tasks we assume that it is rear wheel driven for simplicity.

Task 1.2: Understeer gradient

Load case 1 - Understeer Gradient: 0.00117. Load case 2 - Understeer Gradient: -0.00117. Load case 3 - Understeer Gradient: 0.00027.

Task 1.3: Critical/Characteristic speed

Load case 1 - Critical Speed: 0.00 m/s, Characteristic Speed: 47.80 m/s. Load case 2 - Critical Speed: 47.80 m/s, Characteristic Speed: 0.00 m/s. Load case 3 - Critical Speed: 0.00 m/s, Characteristic Speed: 100.18 m/s.

Task 1.4: Steering wheel angle for one certain operating point

Load case 1 - Steering Wheel Angle: 16.90 degrees. Load case 2 - Steering Wheel Angle: 8.37 degrees.
Load case 3 - Steering Wheel Angle: 13.60 degrees.

Task 1.5 Steering wheel angle for varying operating point

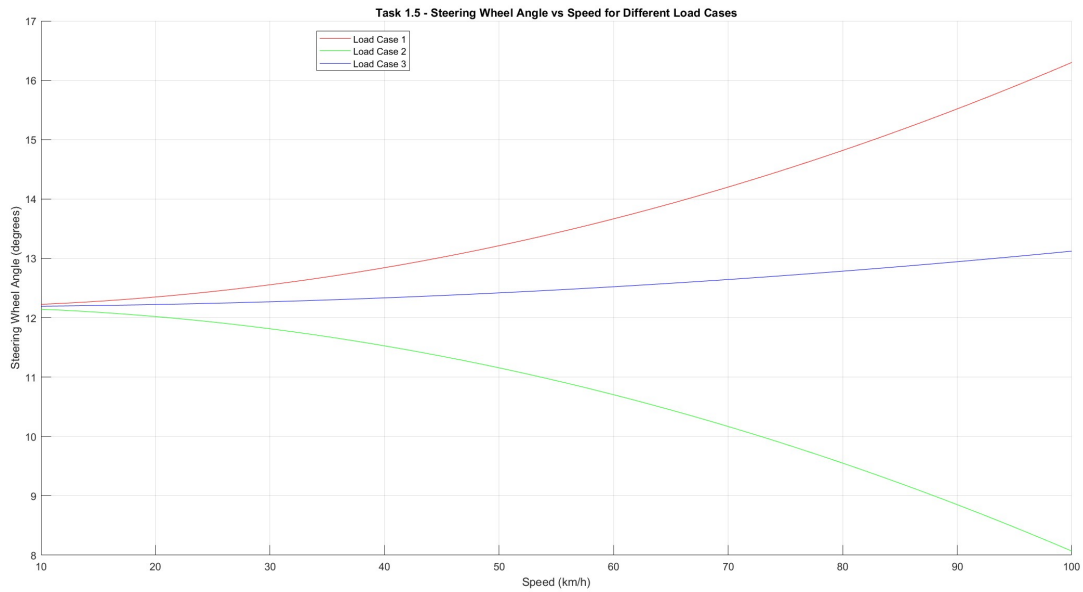


Figure 7: Steering angle vs Speed for different loadcases

Load case 1 is Understeer, Load case 2 is Oversteer, Load case 3 is Neutral steer

Task 2 - Simulation with single track model

Task 2.1: Implement model

Task 2.2: Verify model for moderate a_y

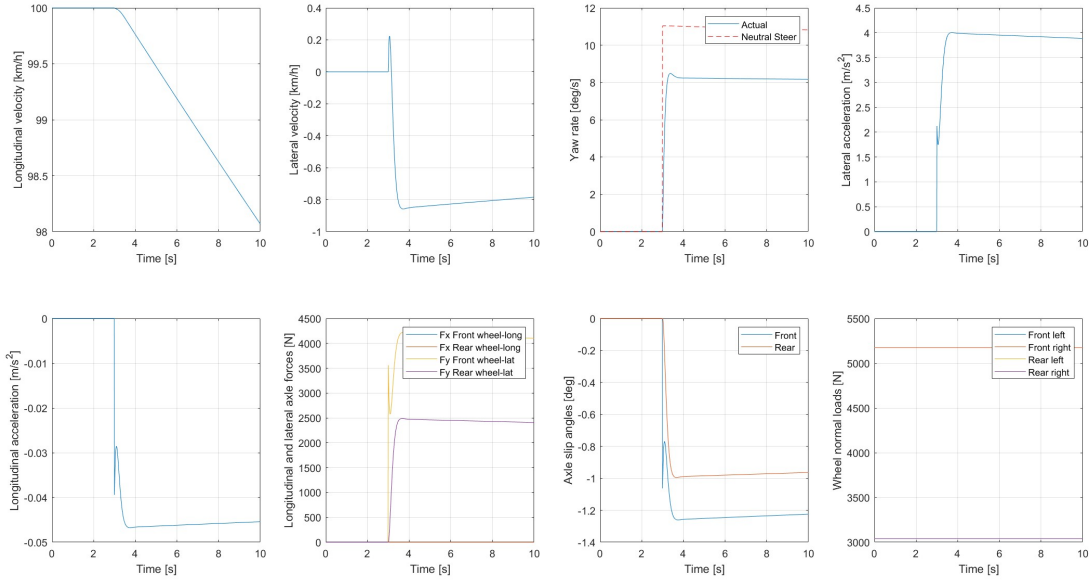


Figure 8: Task2b - Loadcase 1 - 0.37 lf

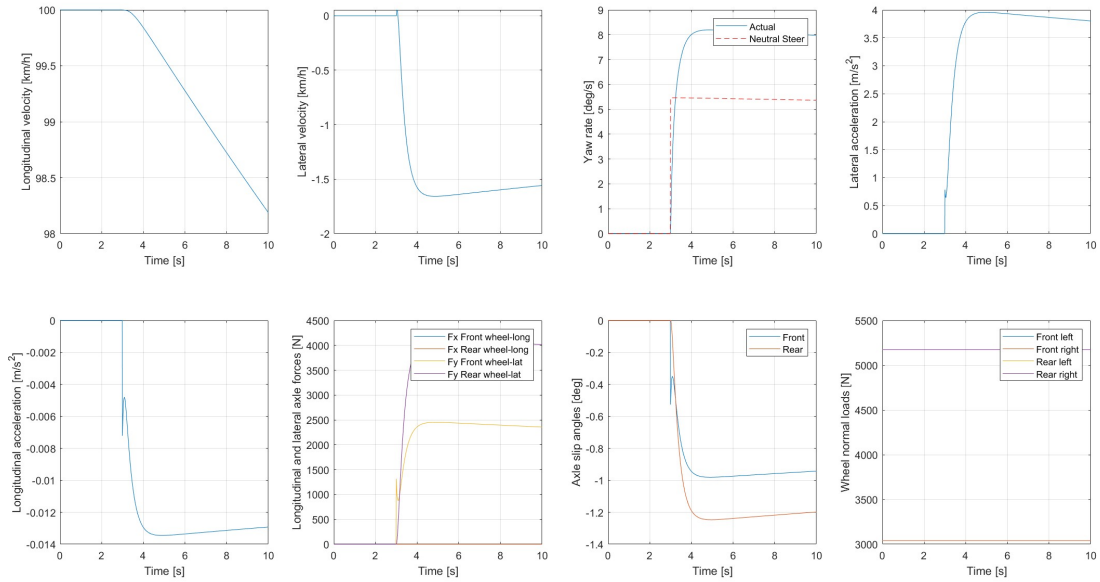


Figure 9: Task2b - Loadcase 2 - 0.63 lf

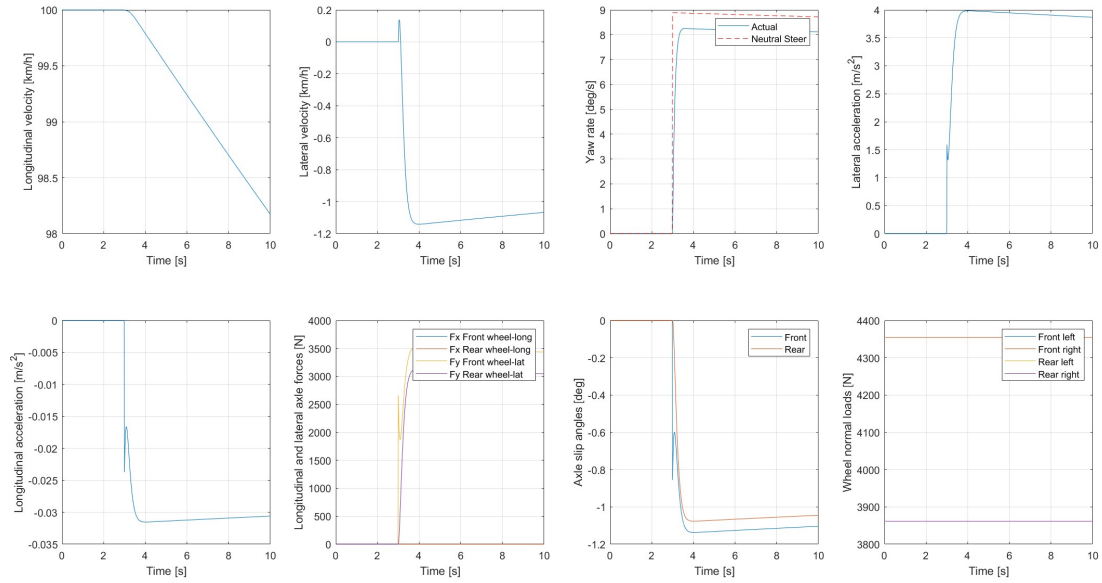


Figure 10: Task2b - Loadcase 3 - 0.47 lf

Yes it does reach $4m/s^2$ lateral acc in all the cases

Task 2.3: Transient response

Load case 1 - Understeer vehicle setup will reach $4m/s^2$ lateral acc in 3.72 sec respond quickest to steering input

Task 2.4: Try model at over-critical speed

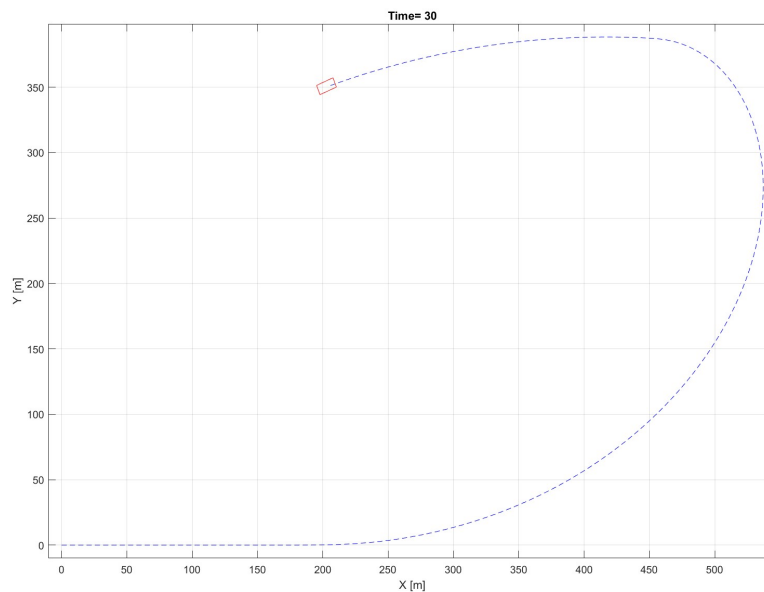


Figure 11: Behaviour over critical speed for a steering angle of 3 degrees

Yup it's oversteering

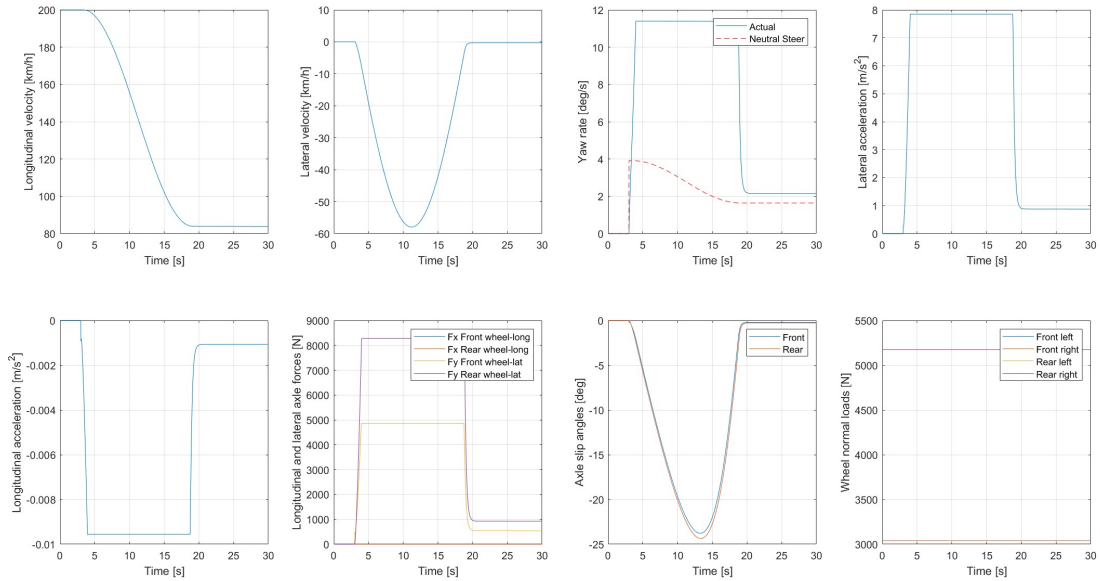


Figure 12: Behaviour over critical speed for a steering angle of 3 degrees

No, the results does not match with the Task 1 because we are increasing the initial speed 55.55 m/s (200km/h) above the critical speed 47.80 m/s, where yaw velocity grows largely with a small steering input to the vehicle.

Task 3 - Load transfer

Task 3.1: Add load transfer to model

Task 3.2: Influence from load transfer on steering response

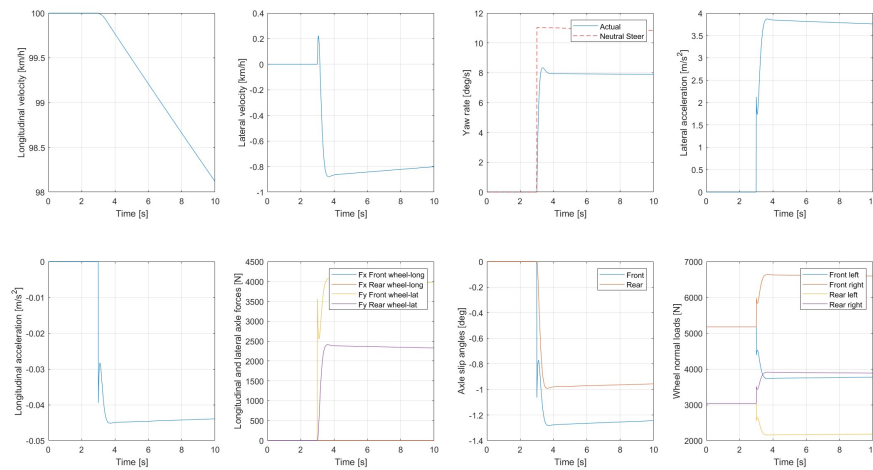


Figure 13: Task3b - Loadcase 1 - 0.37 lf - 0.65 roll stiffness

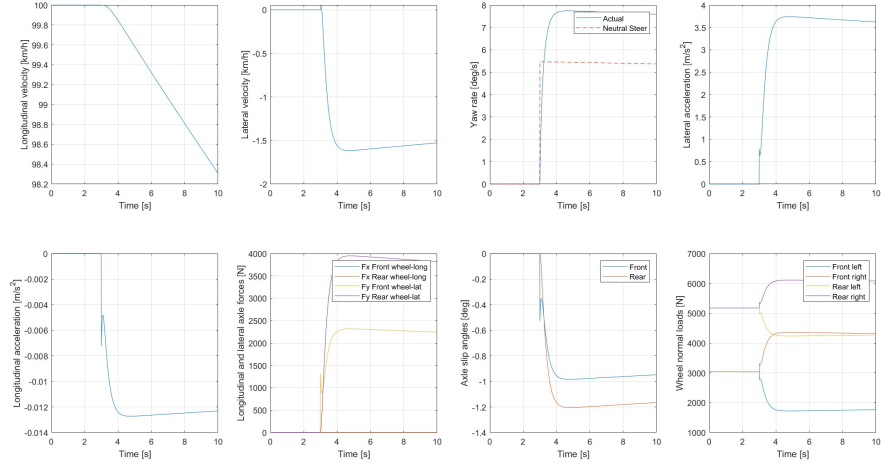


Figure 14: Task3b - Loadcase 2 - 0.63 lf - 0.65 roll stiffness

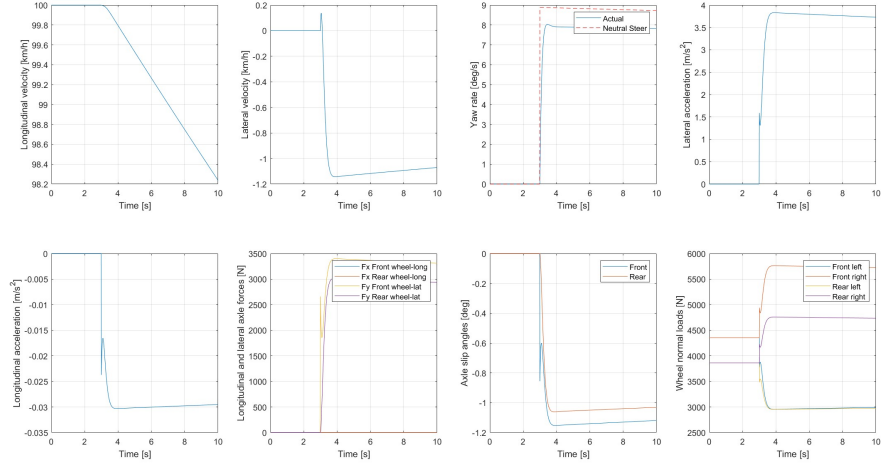
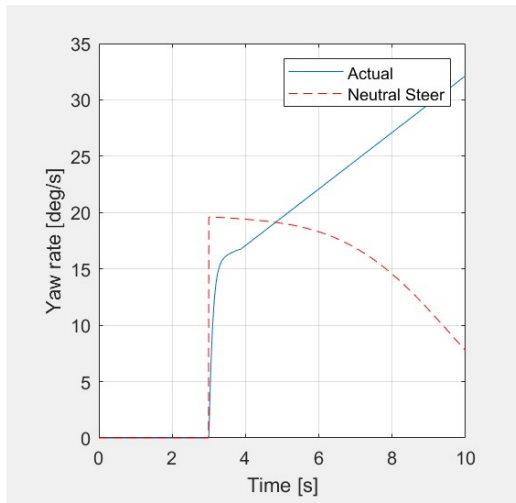


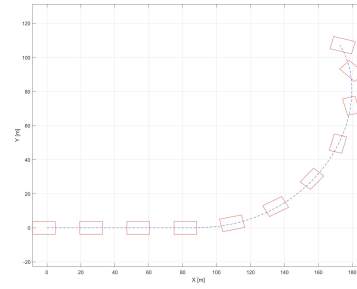
Figure 15: Task3b - Loadcase 3 - 0.47 lf - 0.65 roll stiffness

In Load case 1 the lateral acc is 3.87 m/s^2 , percentage change is 3.25, Load case 2, the lat acc is 3.72 m/s^2 , percentage change is 7, Load case 3, the lat acc is 3.82 m/s^2 , percentage change is 4.5. The load transfer equation is added which means centre of gravity is acting and also the weight disturbance on the axle.

Task 3.3: Influence from roll stiffness distribution on steering response

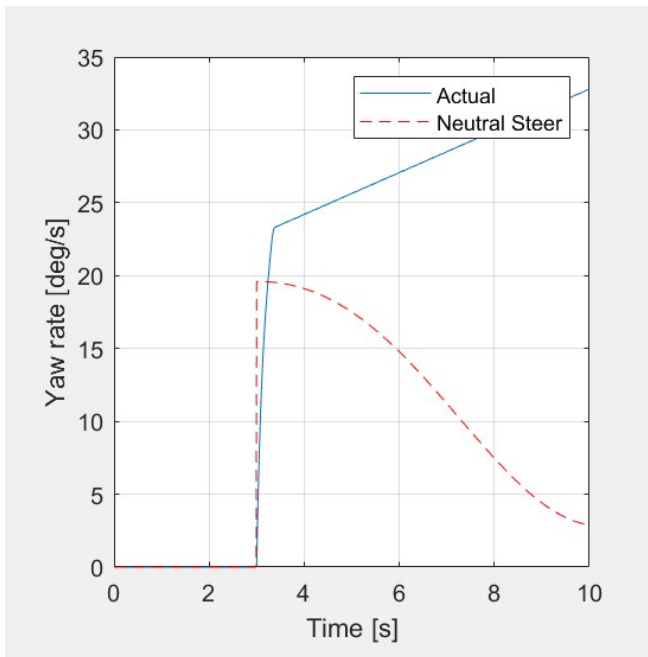


(a) Task3b - Loadcase 1 - 0.47 lf - 0.45 roll stiffness

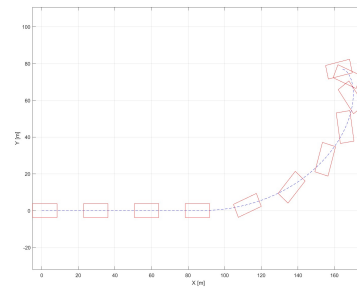


(b) Task3b - Loadcase 1 - 0.37 lf - 0.45 roll stiffness

Figure 16: Comparison for loadcase 1 configurations

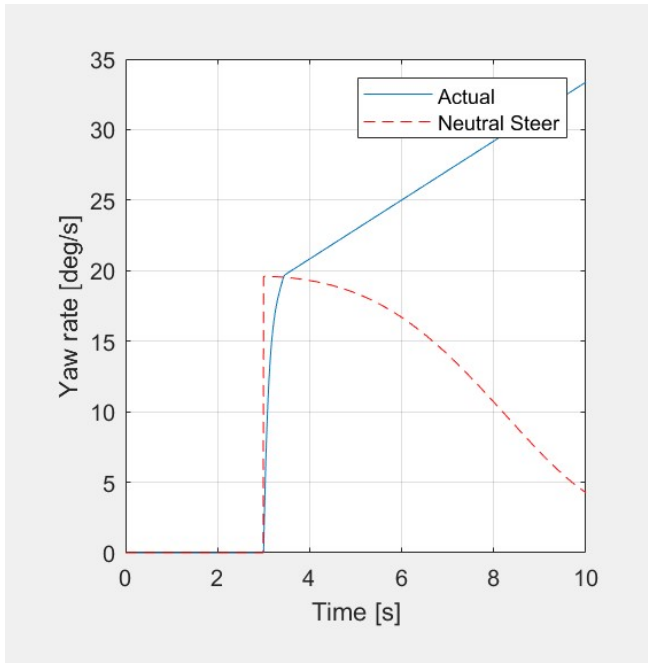


(a) Task3b - Loadcase 2 - 0.63 lf - 0.45 roll stiffness

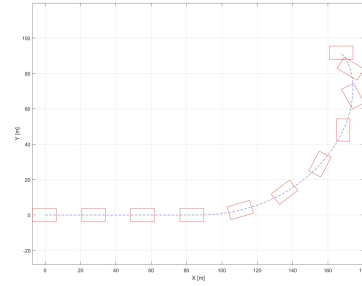


(b) Task3b - Loadcase 2 - 0.63 lf - 0.45 roll stiffness

Figure 17: Comparison for loadcase 2 configurations



(a) Task3b - Loadcase 3 - 0.47 lf - 0.45 roll stiffness



(b) Task3b - Loadcase 3 - 0.47 lf - 0.45 roll stiffness

Figure 18: Comparison for loadcase 3 configurations

In load case 3 When we change the roll stiffness distribution from 0.65 to 0.45 we can see that there is oversteer and more load is at the rear end of the vehicle.

Task 3.4: Propose roll stiffness distribution

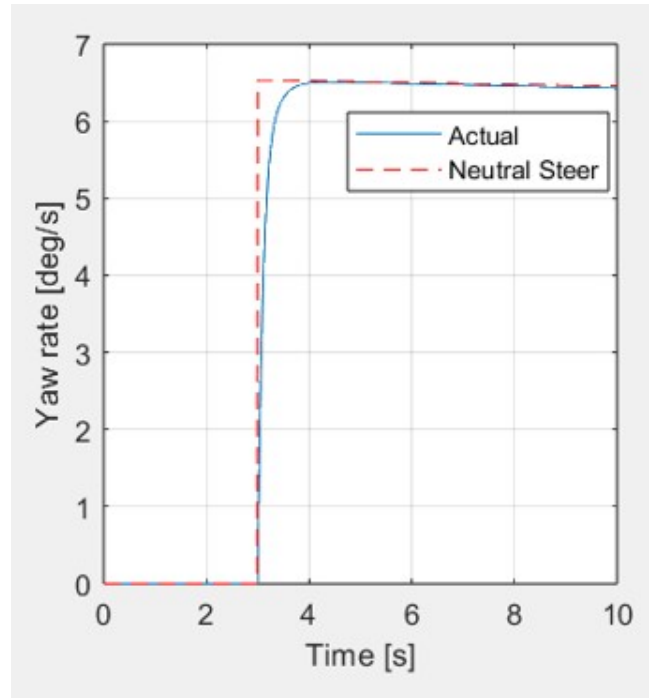


Figure 19: Task3.4 - Loadcase 3 configuration

0.24 roll stiffness achieves neutral steer so thats good. If the production car is designed with one load case we can use the obtained roll stiffness distribution, but the in reality the production car is designed for diffeerent load cases so we do not recommend to use the obtained the roll stifness distribution.

Task 4 - Combined tyre slip

Task 4.1: Add combined slip to model

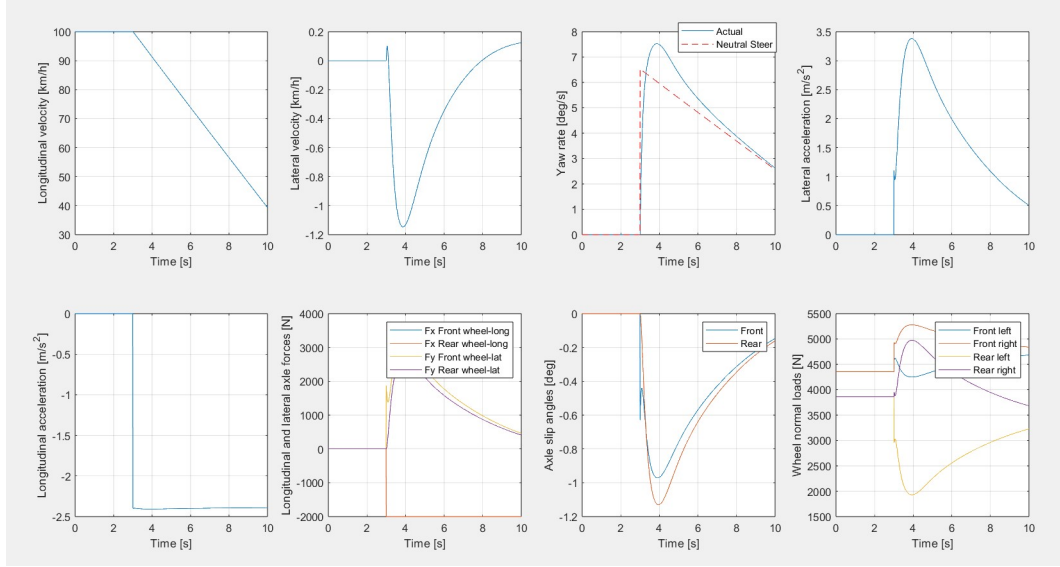


Figure 20: Task 4.1 - Loadcase 3 configuration

No the vehicle does not neutral steer because we have implemented the combined slip tyre model due to which car deviate from the understeer due to reduced lateral forces at the tyres.

Task 4.2: Influence of brake force distribution

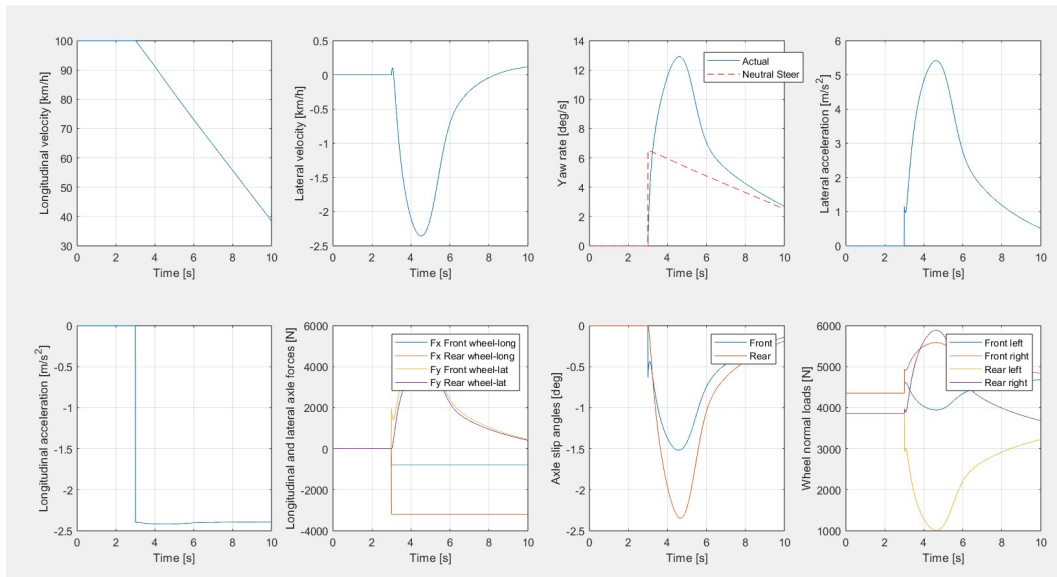


Figure 21: Task 4.2 - Loadcase 3 configuration

When 20% of brake distribution is done on the front axle the vehicle deviates from neutral steer to oversteer and because more load is transferred to the rear end losing grip on the rear end.

Task 4.3: Propose brake force distribution

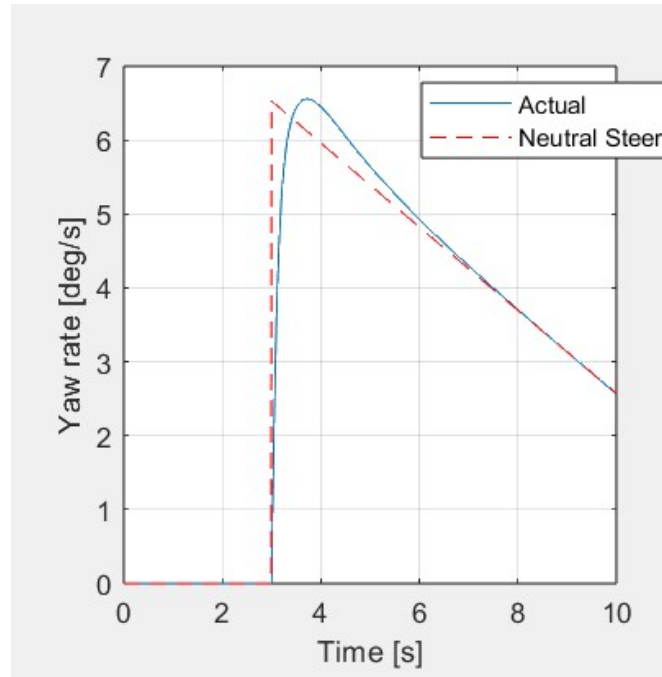


Figure 22: Task4.3 - Loadcase 3 configuration

At 0.69 brake force distribution the vehicle is neutral steered while braking. Yes the determined brake force distribution makes the vehicle neutral steer under braking irrespective of the brake demand but slightly it might under or over steer under extreme condition.

Task 5 - Driving Experience in Simulator

Task 5.1: Participate in model validation session

Parameters	SWA [deg]	Brake Force [N]	Brake Distribution Ratio [-]	Roll Stiffness Distribution [-]	Speed [kph]	If/L	Longitudinal acc		Lateral acc		Yaw velocity	
							max	min	max	min	max	min
Load Case 1	10	0	0.5	0.65	80	0.47	-0.01282	0	1.91	0	4.94	0
Load Case 2	0	2000	0.5	0.65	80	0.47	-1.194	0	0	0	0	0
Load Case 3	3	1000	0.5	0.65	80	0.47	-0.598	0	0.57	0	1.5	0

Figure 23: Max and Min values for driving simulator

For 3 Load cases we have got the max and min values for the longitudinal acc, lateral acc and yaw velocity. The values we got were pretty close with the values of the instructor. For longitudinal acc the values are written in negative because graph is going down in negative so it is mentioned with the negative sign.

Task 5.2: Participate in the simulator drive session

We have participated in simulator drive session and drove the car in certain conditions.

1. Steady state Skid-pad - In this session we drove the car in the constant radius and at slowly increasing speed phase. At 80/20 weight distribution, more weight was on the front side, we felt heavy steering and the car was under steering and it was difficult to make a turn at increasing speed. At 20/80 weight distribution the steering felt lighter and the car was oversteering and for a small steering input the car was off the track at higher speed.
2. Over-speeding into the curve - In this session we drove the car at 90km/hr speed before entering the curve, hit a brake, and take a curve. At 57/43 brake bias, which feels like real car set up more stability during cornering. At 80/20 brake bias setup the car caused significant understeer, and it was difficult to make a turn. At 20/80 brake bias setup, the car caused oversteer, making the car spin.
3. Model comparison with Caster Vehicle Model - The Porsche vehicle model setup seems, moreover, reality and slight oversteer because of its setup for the track event. The car was stable, the grip on the tires was good and many other aspects were taken into account when we compared it to our model.

Task 5.3: Documentation in assignment report

Some of the additional aspects listed below to be considered when integrating the simulation model in the simulator model.

1. - Simulator - The consistent input/output interface between the simulator and the model to avoid data exchange issues. The simulator allows for the parameter adjustment to accommodate variation in the simulator model.
2. - Visualization - Interactive visualization to observe the model behavior during run-time.
3. - Real-Time application - Simulation time is integrated with real- time hardware for precise control and data acquisition.

From the log data from steady-state constant radius test we have plotted graphs below steering angle vs lateral acceleration.

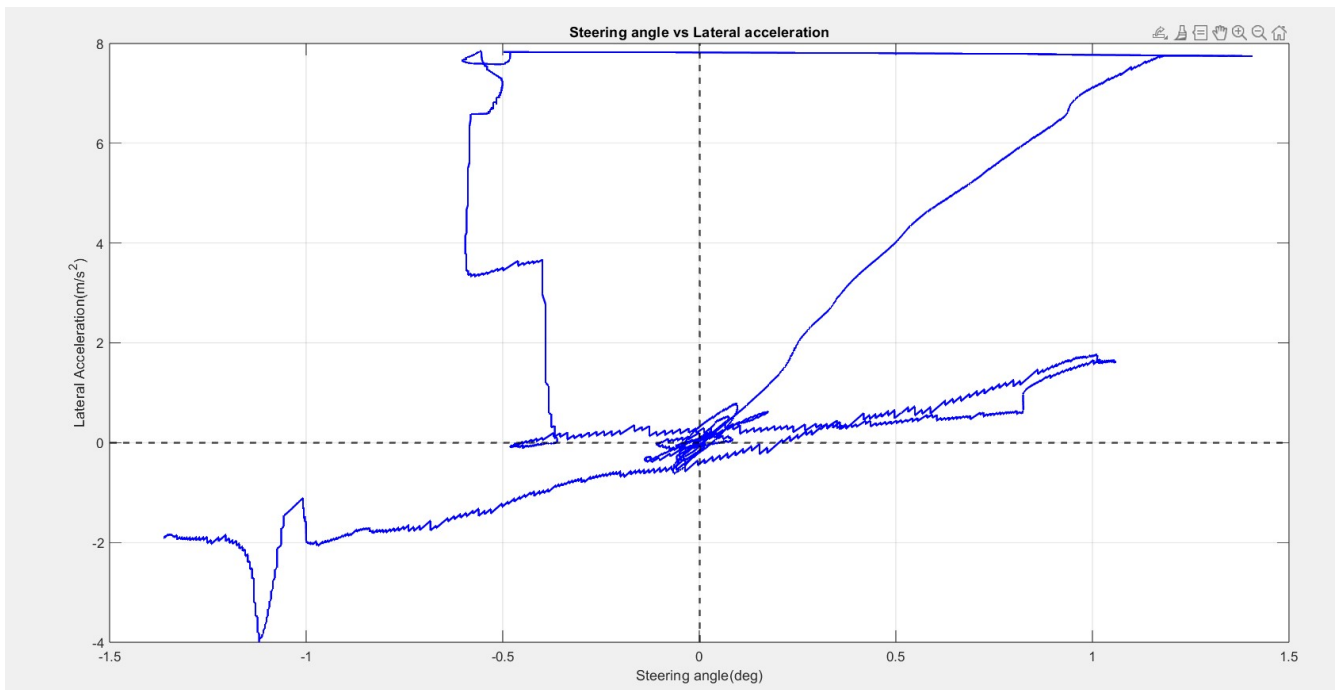


Figure 24: Brake Test 1

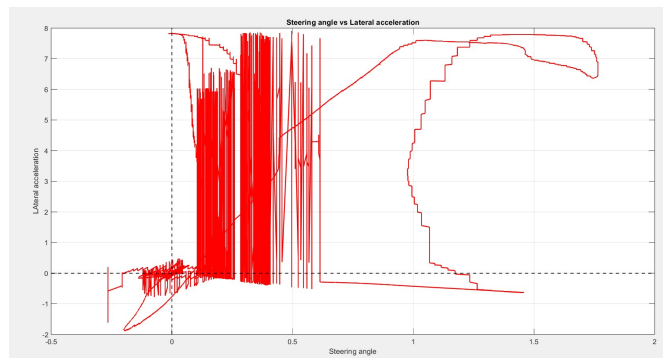


Figure 25: Brake Test 2

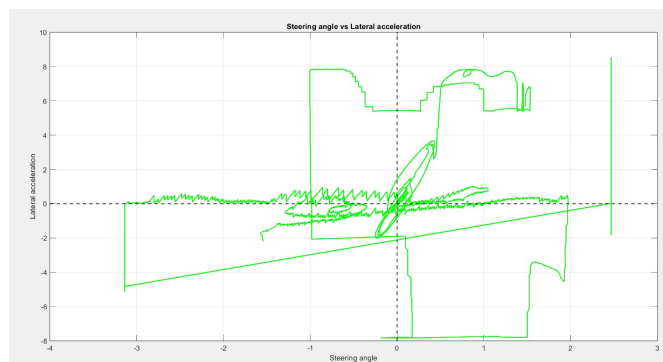


Figure 26: Brake Test 3

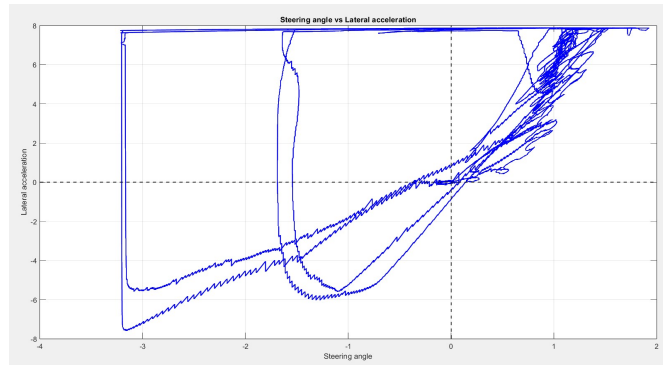


Figure 27: Steady state skid pad 1

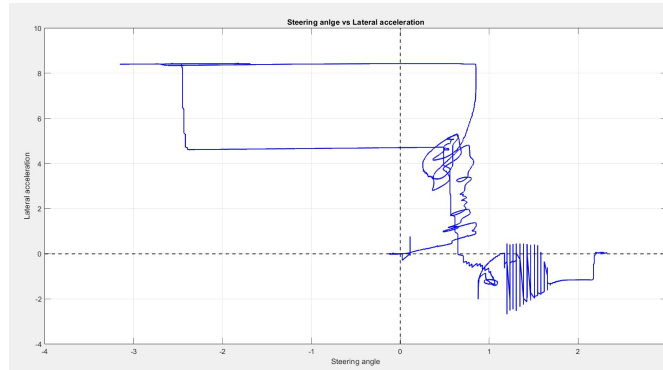


Figure 28: Steady state skid pad 2

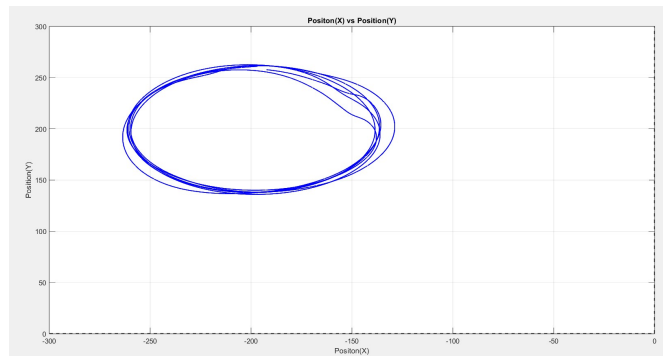


Figure 29: Steady state skid pad 1: Position of the car

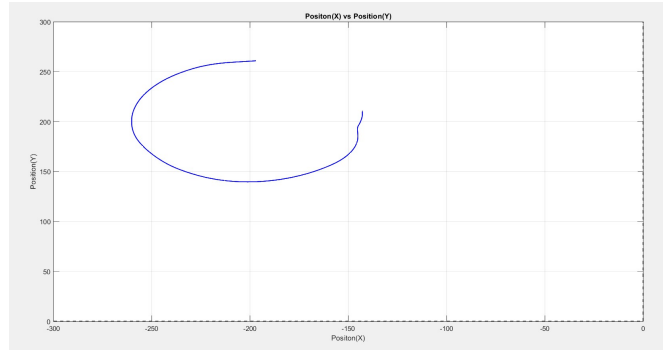


Figure 30: Steady state skid pad 2: position of the car

From the log data the above graphs are plotted, and we can say that there is loose grip and its hard to predict whether its under-steer or over-steer for a certain weight distribution when we compare it to the handling diagrams. From Fig. 29, the Skid pad 1 by seeing the position of the car we can say that the car is understeering and the weight distruibution is 80/20. From fig 30, skid pad 2 the position of the car we can say that the car is oversteering as it discontinues from the path and the weight distribution is 20/80.